1 **(a)** Translation **[B1]**

by vector $\begin{pmatrix}4\\0\end{pmatrix}$ **[B1]**

(b) $f(x)+7$ **[B1]**

[Total Marks: 3]

2 **(a)** x^2-5 **[B1]**

(b) $(x-3)^2$ **[B1]**

(c) Translation **[B1]**

by vector $\begin{pmatrix}-4\\-1\end{pmatrix}$ **[B1]**

[Total Marks: 4]

3 **(a)** $g(x)=f(x)+7$ **[M1]**

$g(x)=2x+2$ **[A1]**

(b) Translation **[B1]**

by vector $\begin{pmatrix}4\\0\end{pmatrix}$ **[B1]**

[Total Marks: 4]

4 **(a)** Translation **[B1]**

By vector $\begin{pmatrix}3\\4\end{pmatrix}$ **[B1]**

(b) $f(x-3)+4$ **[B2]**

(Award [B1] for $f(x-3)$ or $+4$ seen)

(c) $(-3,-3)$ **[B1]**

[Total Marks: 5]

5 **(a)** Horizontal shift 3 seen **[B1]**

$\begin{pmatrix}3\\-2\end{pmatrix}$ **[B1]**

(b) $(x-3)^2$ seen **[B1]**

$(x-3)^2-2$ **[B1]**

(c) $x=-5$ **[B1]**

$(-5,-1)$ **[B1]**

[Total Marks: 6]

6 **(a)** $(-2, 1)$ **[B1]**

(b) $(5, -1)$ **[B1]**

(c) **(i)** $a = 2$ **[B1]**

$b = -5$ **[B1]**

(ii) $(5, -6)$ **[B1]**

(Strict follow through from their vector in (c)(i) applied to $(3, -1)$)

[Total Marks: 5]

7 **(a)** Writes $g(x) = f(x-2) - 5$ or $(x-2)^2 - 4(x-2) + 1 - 5$ **[M1]**

Expands correctly: $(x^2 - 4x + 4) - 4x + 8 - 4$ **[M1]**

$x^2 - 8x + 8$ **[A1]**

(b) Completes the square: $f(x) = (x-2)^2 - 3$ and $h(x) = (x+1)^2 - 3$ **[M1]**

Identifies horizontal shift of -3 units **[A1]**

$\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ **[A1]**

[Total Marks: 6]

8 **(a)** $g(x) = f(x+2) = 3(x+2)^2$ **[M1]**

$3(x^2 + 4x + 4)$ **[M1]**

$3x^2 + 12x + 12$ **[A1]**

(b) $(-2, -8)$ **[B2]**

(Allow B1 for each correct coordinate)

(c) Substitutes $x = 0$ into $h(x) = 3(x+2)^2 - 8$ (or similar valid method) **[M1]**

4 **[A1]**

[Total Marks: 6]

9 **(a)** Translation by 2 units left or 3 units down **[M1]**

Correct line segments joining $(-3, -2)$, $(-1, 1)$, $(1, -2)$, and $(2, -1)$ **[A1]**

(b) Translation **[B1]**

Vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ **[B1]**

(c) $f(x) - 5$ **[B1]**

(d) $x = 4 + 3 = 7$ or $y = 2 + 4 = 6$ **[M1]**

$(7, 6)$ **[A1]**

[Total Marks: 7]

10 **(a)** $(4, -1)$ **[B1]**

(b) $f(x+3)+2$ **[B2]**

(B1 for $f(x+3)$ and B1 for $+2$)

(c) $f(-k) = 0$, so $-k = 2$ or $-k = 6$ **[M1]**

$k = -2$ and $k = -6$ **[A1]**

(d) $f(x-3)$ **[B1]**

[Total Marks: 6]

11 **(a)** Correct vertical shift applied: $(x^2 - 4) + 5$ **[M1]**

$x^2 + 1$ **[A1]**

(b) $(-3, -4)$ **[B2]**

If 0 scored, award 1 mark for x-coordinate $= -3$ or y-coordinate $= -4$ **[B1]**

[Total Marks: 4]

12 **(a)** Translation **[B1]**

by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ **[B1]**

(b) Recognising that $g(x) = f(x-4)$ and substituting: $(x-4)^3 - 3(x-4)$ **[M1]**

Expanding $(x-4)^3 = x^3 - 12x^2 + 48x - 64$ **[B2]**

If 0 scored for cubic expansion, award 1 mark for $(x-4)(x^2 - 8x + 16)$ **[M1]**

$x^3 - 12x^2 + 45x - 52$ **[A1]**

[Total Marks: 6]

13 **(a)** $g(x) = f(x+4)$ **[M1]**

$(x+4)^2 - 2(x+4) - 3$ **[M1]**

$x^2 + 8x + 16 - 2x - 8 - 3$ **[M1]**

$x^2 + 6x + 5$ **[A1]**

(b) $h(x) = x^2 - 2x - 3 - 5 = x^2 - 2x - 8$ **[M1]**

$(x-4)(x+2) = 0$ **[M1]**

$x = 4$ and $x = -2$ **[A1]**

[Total Marks: 7]

14 **(a)** Local maximum plotted at $(1, 4)$ **[B1]**

Local minimum plotted at $(3, 0)$ and y-intercept or key point $(2, 2)$ correct **[B1]**

Smooth cubic curve drawn through translated points with correct shape **[B1]**

(b) x-coordinate $= 1 - k$ **[B1]**

y-coordinate $= -1 - 2 = -3$, giving $(1 - k, -3)$ **[B1]**

(c) $p = -3$ **[B1]**

$p = 1$ **[B1]**

[Total Marks: 7]

15 (a) $2(x^2 - 4x) + 11$ or $a = 2$ **[B1]**

$2[(x - 2)^2 - 4] + 11$ or $h = -2$ **[B1]**

$2(x - 2)^2 + 3$ **[A1]**

(b) Completing the square for $g(x)$: $2(x + 1)^2 - 5$ OR equating $f(x - p) + q = 2x^2 + 4x - 3$ **[M1]**

Identifying the minimum point of $f(x)$ as $(2, 3)$ and the minimum point of $g(x)$ as $(-1, -5)$ **[M1]**

Horizontal shift $= -1 - 2 = -3$ **[A1]**

Vertical shift $= -5 - 3 = -8$, giving vector $\begin{pmatrix} -3 \\ -8 \end{pmatrix}$ **[A1]**

[Total Marks: 7]

16 (a) Finds $(x - 3)^2$ as part of completing the square: $f(x) = (x - 3)^2 - 9 + 14$ **[M1]**

$f(x) = (x - 3)^2 + 5$ **[M1]**

Translation vector $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ **[A1]**

(b) Substitutes $x - 4$ into $g(x)$: $g(x - 4) = \dfrac{3}{(x - 4) + 2} - 5$ **[M1]**

Adds 2 to the function: $h(x) = \dfrac{3}{x - 2} - 5 + 2$ **[M1]**

$h(x) = \dfrac{3}{x - 2} - 3$ or equivalent single fraction $\dfrac{9 - 3x}{x - 2}$ **[A1]**

(c) $x = 2$ **[B1]**

$y = -3$ **[B1]**

(Strict follow through from their $h(x)$ in part (b))

[Total Marks: 8]

17 (a) (i) $\dfrac{20}{x} - 4$ **[B1]**

(ii) $\dfrac{20}{x} - 4 = 0$ **[M1]**

$x = 5$ **[A1]**

(b) $h(x) = f(x - k) = \dfrac{20}{x - k}$ **[M1]**

$\frac{20}{8-k} = 5$ **[M1]**

$k = 4$ **[A1]**

(c) Vertical asymptote: $x = 3$ **[B1]**

Horizontal asymptote: $y = 7$ **[B1]**

[Total Marks: 8]

18 (a) (i) $f(x+2) - 6$ **[B1]**

(ii) $2(x+2)^2 - 5(x+2) + 1 - 6$ **[M1]**

$2(x^2 + 4x + 4) - 5x - 15$ **[M1]**

$2x^2 + 3x - 7$ **[A1]**

(b) $2(x-p)^2 - 5(x-p) + 1 + q = 2x^2 - 13x + 18$ **[M1]**

$-4p - 5 = -13$ giving $p = 2$ **[A1]**

$2(2)^2 + 5(2) + 1 + q = 18$ giving $q = -1$ **[M1]**

$\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ **[A1]**

(c) $2x^2 - 5x + 1 = 2x^2 + 3x - 7$ **[M1]**

$8x = 8$ giving $x = 1$ **[A1]**

$(1, -2)$ **[A1]**

[Total Marks: 11]